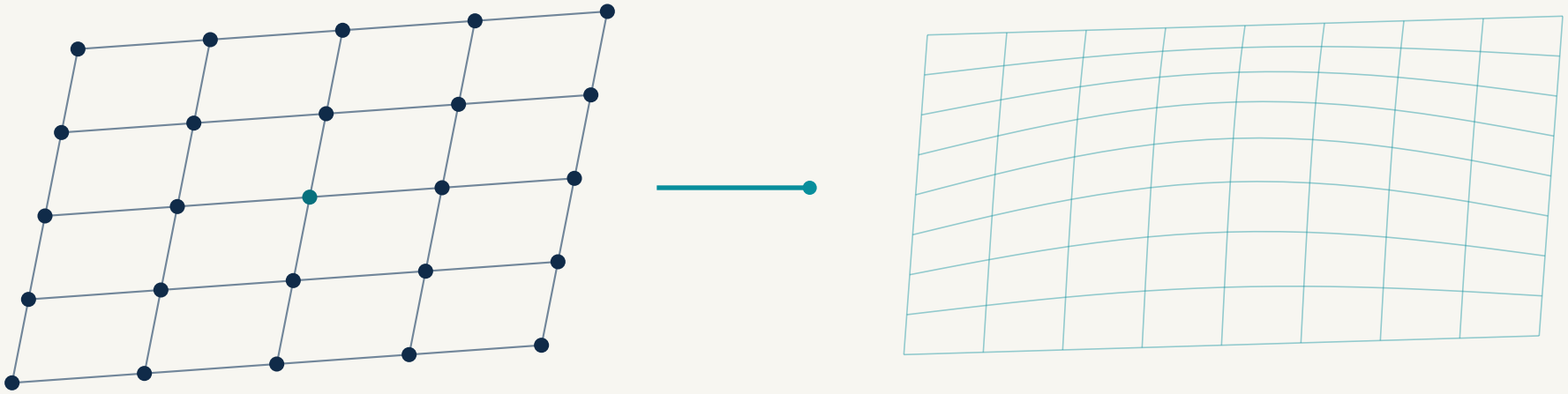
B-Splines for 3D Object Representation

A smooth surface is controlled by a grid of points in three-dimensional space.

A B-spline surface combines a control net with basis functions in two parameter directions. The net guides the shape; the evaluated surface is usually smooth and does not pass through every interior control point.



CONTROL NET P(i,j)

EVALUATED SURFACE S(u,v)

Key idea

Move selected points to change the surface locally.

Use the method for a tabletop, shell, panel, or other freeform part.

u, v in [0, 1]

Method adapted from the supplied tabletop reference slides.

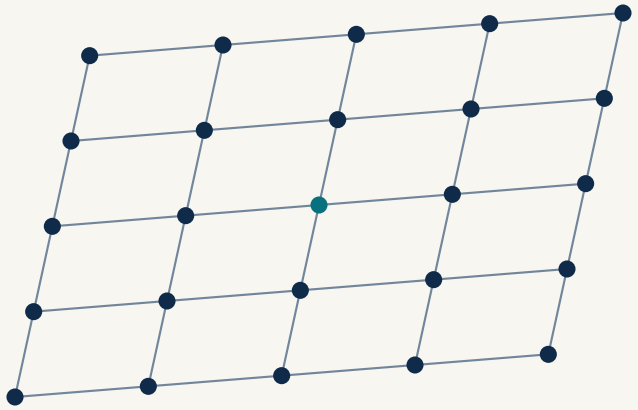
Tensor-Product Surface

Two curve directions combine to define one surface.

$$S(u,v) = \sum_{i=0}^{n} \sum_{j=0}^{m} N_{i,p}(u) \quad N_{j,q}(v) \quad P_{i,j}$$

$$P_{i,j} = (x_{i,j}, y_{i,j}, z_{i,j})$$

Each control point contributes according to the product of its u- and v-direction basis weights.

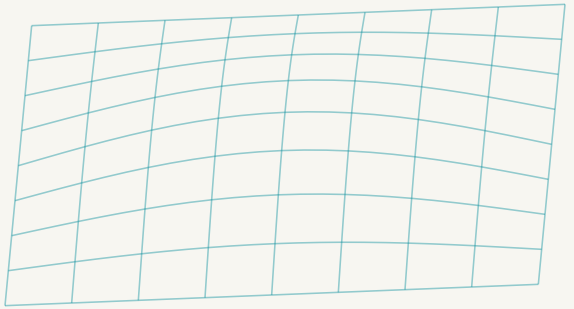


BASIS FUNCTIONS: COX-DE BOOR RECURSION

N(i,0)(u) = 1 when u(i) <= u < u(i+1); otherwise 0

N(i,p)(u) = ((u-u(i))/(u(i+p)-u(i))) N(i,p-1)(u)
+ ((u(i+p+1)-u)/(u(i+p+1)-u(i+1))) N(i+1,p-1)(u)

A zero denominator gives a zero term. Use the final span convention at the right endpoint so the surface includes u = 1 and v = 1.



Weights act in both directions.

Degree, Knots, and Smoothness

Degree shapes each basis function; knots partition the parameter domain.

CUBIC OPEN-UNIFORM EXAMPLE

p = q = 3 U = V = [0, 0, 0, 0, 1/3, 2/3, 1, 1, 1, 1]



Open ends repeat p + 1 times

Clamping makes each boundary meet the first or last control row. Five control points in each direction with this knot vector create three spans.

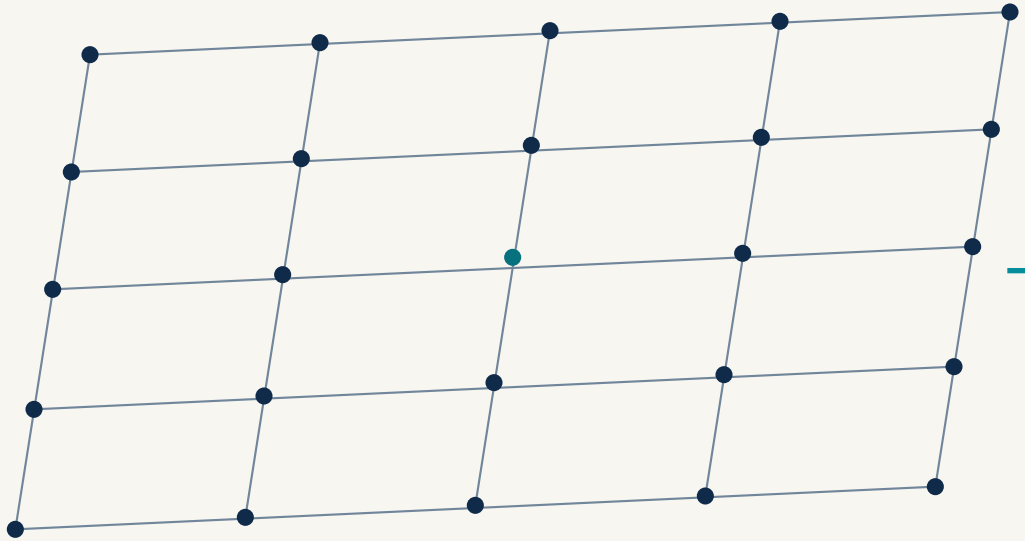
WHAT THE CHOICES DO

- Higher degree gives broader, smoother basis functions.
- A knot marks where a polynomial span ends and the next begins.
- At a knot of multiplicity m, continuity is generally C^(p-m).
- A simple interior knot in a cubic spline gives C2 continuity.

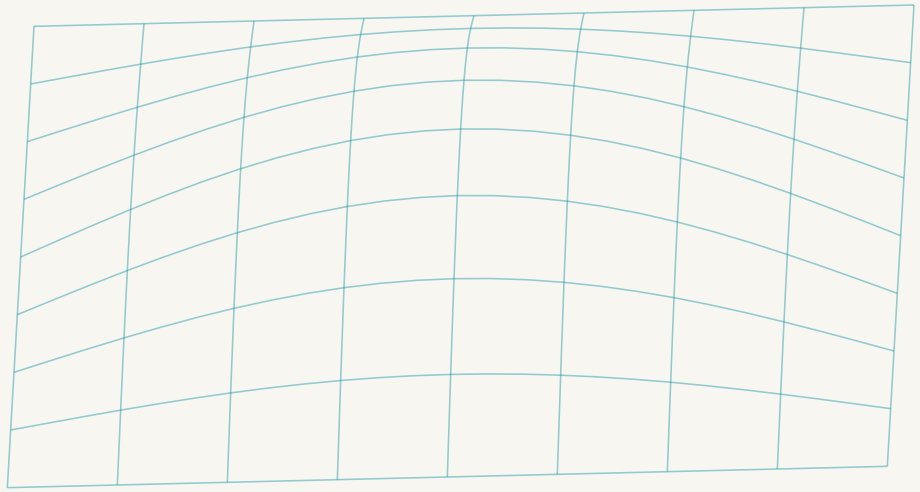
Basis functions are nonnegative and sum to one over the active domain.

How the Control Net Shapes a Surface

Separate x-y placement from z-height to design the geometry.



CONTROL-POINT NET



SMOOTH SURFACE

Edit

One row or point at a time.

Evaluate

Sample S(u,v) over a grid to render the patch.

x, y set the footprint; z offsets add crown, dish, or twist.

B-Splines in a 3D Modeling Workflow

Use a freeform patch where curvature matters, and solids for regular parts.

01

Define

Arrange surface control points.

02

Choose

Set degree and knot vectors.

03

Evaluate

Sample S(u,v) over the patch.

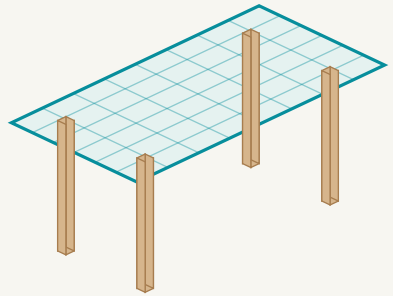
04

Assemble

Join the patch and leg solids.

TABLETOP

B-spline surface with mild crown.



LEGS

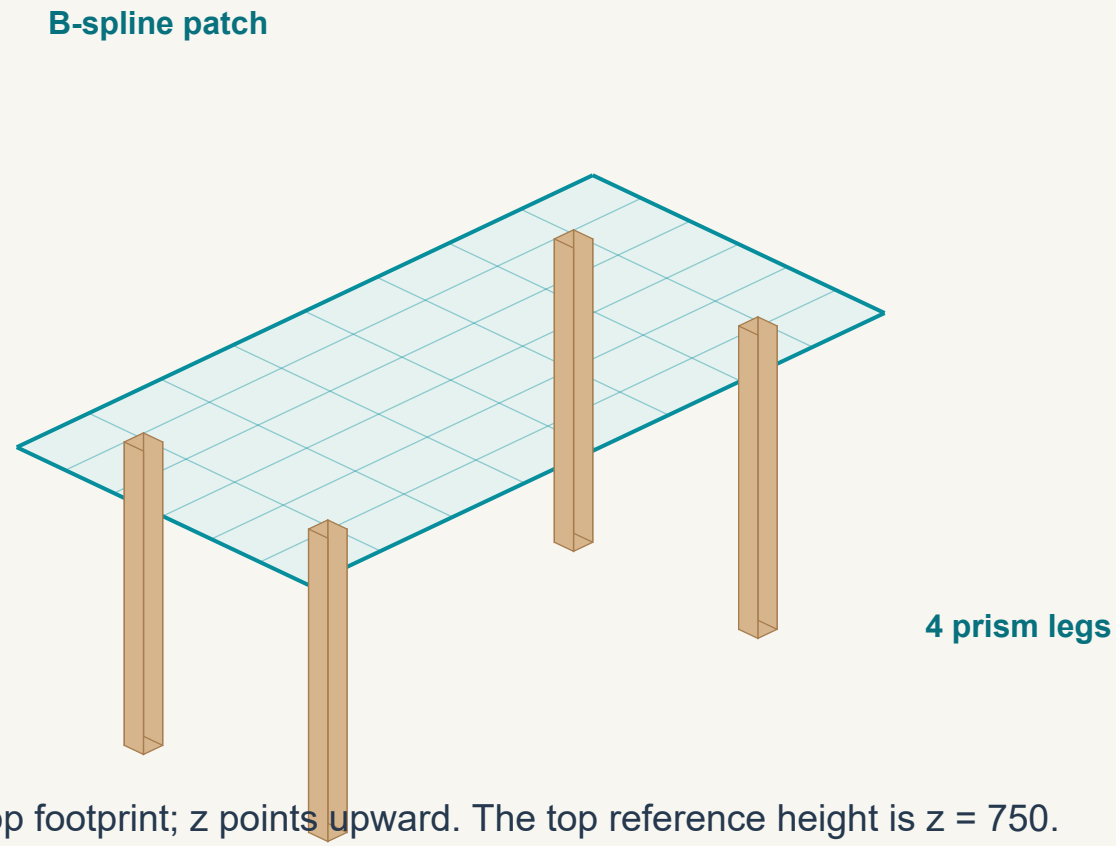
Four rectangular prisms.

Use NURBS when exact circles or conics are required.

Problem: Model a Modern Table

One cubic B-spline tabletop patch and four prism supports make the object.

DESIGN TARGETS | mm	
Top edge z	**750**
Maximum height	**769.1**
Top thickness	**40**
Leg section	**60 x 60**
Crown target	**about 20 at center**



Coordinates: origin at plan center on the floor; x and y span the tabletop footprint; z points upward. The top reference height is z = 750.

Specify the 5 x 5 Control Net

A symmetric height field gives the tabletop a shallow crown and preserves its footprint.

	j=0	j=1	j=2	j=3	j=4
i=0	0	0	0	0	0
i=1	0	8	12	8	0
i=2	0	12	25	12	0
i=3	0	8	12	8	0
i=4	0	0	0	0	0

Height offsets h(i,j), in mm

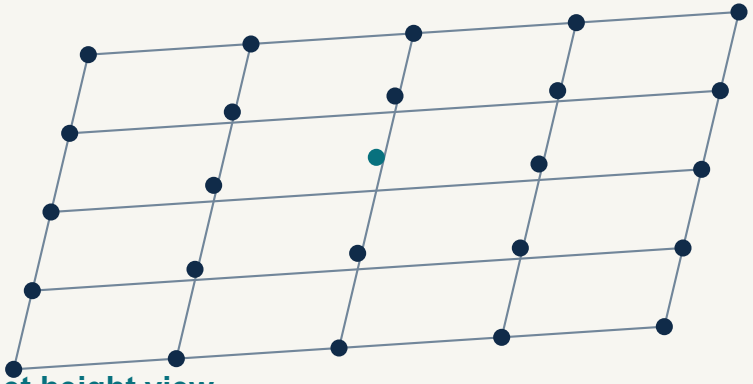
The 5 x 5 grid defines the crown.

Zero perimeter offsets keep the rim level.

P(i,j) = (x(i), y(j), 750 + h(i,j))

x = [-1200, -1200, 0, 1200, 1200] mm

y = [-600, -600, 0, 600, 600] mm



Control-net height view

Evaluate the Center of the Patch

Cubic open-uniform knots in both directions give the center basis weights.

p = q = 3 **U = V = [0, 0, 0, 0, 1/3, 2/3, 1, 1, 1, 1]**

At u = v = 0.5:

[N0, N1, N2, N3, N4] = [0, 1/8, 3/4, 1/8, 0]

Only the central 3 x 3 offsets contribute:

	1/8	3/4	1/8
1/8	8	12	8
3/4	12	25	12
1/8	8	12	8

z offset at center

= [1/8, 3/4, 1/8] H [1/8, 3/4, 1/8]^T

= (1/8)(11) + (3/4)(21.75) + (1/8)(11)

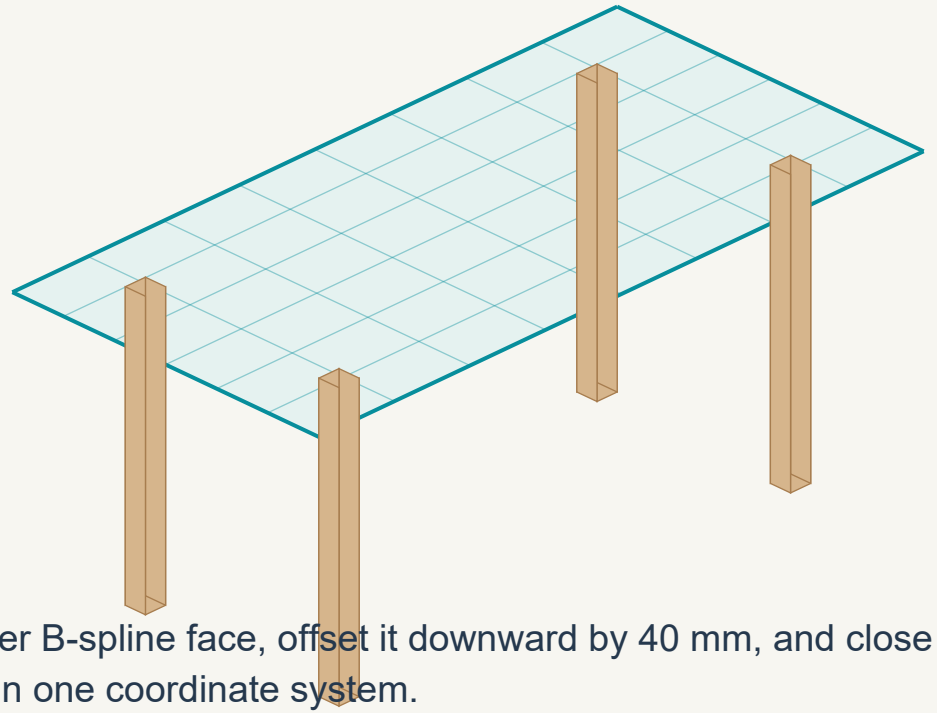
= 19.0625 mm

S(0.5, 0.5) = (0, 0, 769.06) mm

At the center x = y = 0 by symmetry. The 19.06 mm rise is above the reference height z = 750 mm.

Add the Supports and Assemble

Place four 60 mm square legs 250 mm from the long ends and 150 mm from the side edges.



LEG CENTERS | mm

(x, y) = (+/-950, +/-450)

Each support is 60 x 60 mm in section.

Set its length to the local underside height at that position (about 710 mm).

ASSEMBLY TRANSFORM

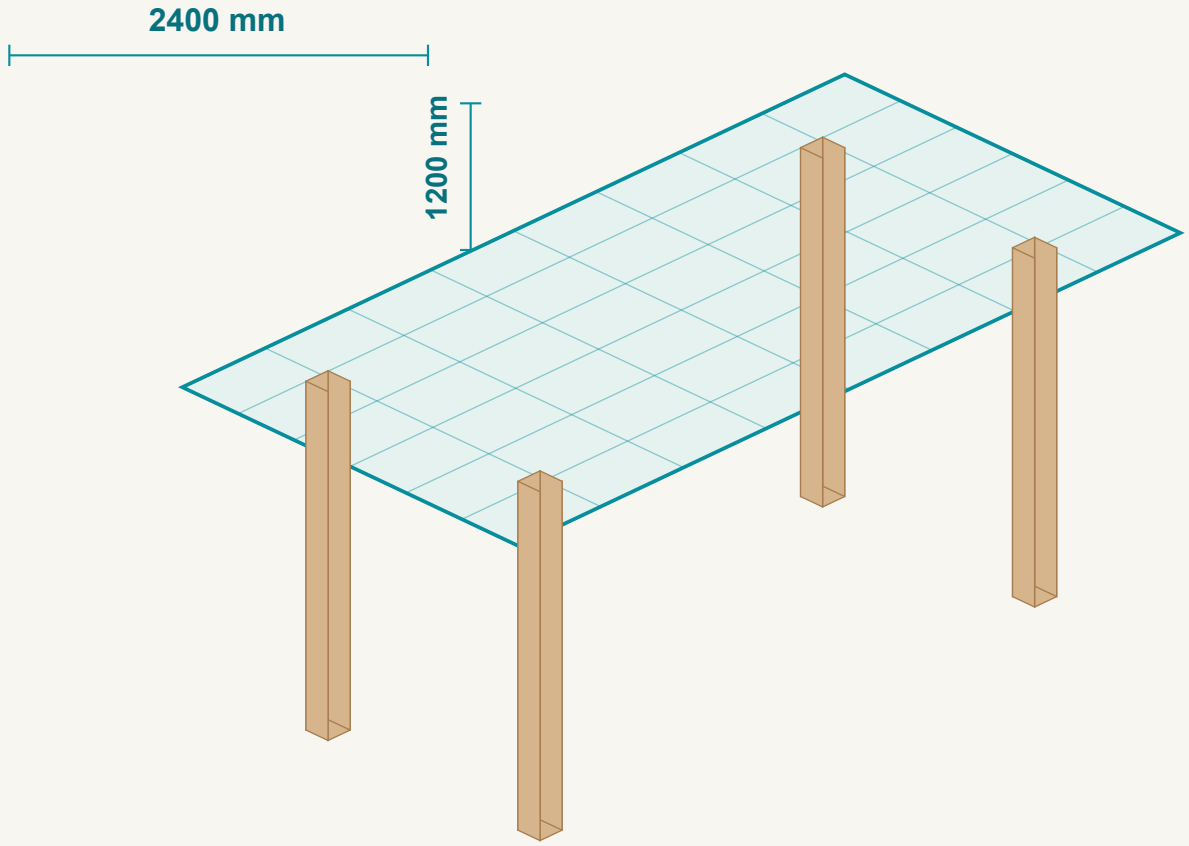
Leg k: local prism + T(xk, yk, 0)

Top shell: surface + 40 mm offset

Build the upper B-spline face, offset it downward by 40 mm, and close the perimeter. Align each leg to the lower face and combine all components in one coordinate system.

Final 3D Table Model

One smooth B-spline patch and four simple solids form the complete object.



MODEL CHECK

- 2400 x 1200 mm footprint
- 750 mm edge height
- 40 mm tabletop thickness
- 769.1 mm peak height
- Four 60 x 60 mm legs